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The Next Phase of AI: From Intelligence to Embodiment

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Back in January, Elon Musk said that Tesla will begin production of “several thousand” Optimus robots by the end of 2025. “It’s one of those things where I think, long term, Optimus has the potential to be north of \$10 trillion in revenue.” Last week, in a two-hour keynote speech at the NVIDIA GTC conference, CEO Jensen Huang made it clear that the next great leap in artificial intelligence isn’t just smarter software — it’s the embodiment of AI into the physical world. “We’re entering the era of robotics,” Huang declared, as he introduced a humanoid robot developed in partnership with Google DeepMind and Disney. For the first time, this shift in AI is being visualized not through lines of code or chatbot conversations, but through walking, sensing, and reasoning machines that can interact with the world in real time. This moment marks a turning point where AI, once trapped behind screens, is being set free to move, see, and act.

The concept of embodied AI extends far beyond humanoid robots. It represents a fundamental shift in computing: one where artificial intelligence is not just a cloud-based model answering questions, but a reasoning agent deployed inside phones, computers, autonomous vehicles, and physical assistants. This transition follows the trajectory Marc Andreessen predicted in his influential 2011 “Software is Eating the World” op-ed, with the Magnificent 7 tech giants (Apple, Microsoft, Google, Amazon, Meta, Nvidia, and Tesla) exemplifying how cloud-based software has dominated the economy for over a decade. The weightings in MSCI World and the SPX today look very different than in 2007 before the rise of the Magnificent 7. Now, we’re at an important inflection point as AI evolves beyond the cloud into embodied systems with physical presence. Just as software companies transformed market indices over the past decade, the rise of embodied AI will likely reshape stock market weightings five years from now, creating new leaders in hardware, robotics, and integrated AI systems. This leap from purely digital to embodied AI parallels the transformation from desktop to mobile internet — it will change how we live, work, and experience the world. Embodied AI will understand context through vision, sound, and touch, unlocking real-world problem-solving abilities and interactive capabilities that were impossible before.

At Tesla AI Day and in subsequent interviews, Musk has envisioned a future where humanoid robots — like Tesla’s Optimus — perform dangerous, boring, or repetitive tasks across manufacturing, logistics, and even home care. While many are focused on the novelty, the economic impact could be profound: robots powered by AI models trained on billions of hours of real-world video and simulated environments could lead to a productivity boom across sectors. Chris Camillo, from the book *Unknown Market Wizards*, said on a podcast recently: “Occasionally there’s one that comes around once in a lifetime... and this is it for me. This is the biggest trade of my life — and it’s the robot trade. It’s — you know — the fancy way to say robots or humanoids. It is embodied AI, it’s the embodiment of artificial intelligence.” As Jensen Huang put it, “Every company will be a robotics company.”

But what truly sets this new era apart is the unprecedented demand for compute. When DeepSeek released R1 on Inauguration Day, it started a psychological shift in investors’ minds about compute and caused doubt about how much was actually going to be needed. For some reason, Huang’s speech did not get as much of investor’s attention from the other side. Unlike traditional machine learning models that respond to static prompts, embodied AI requires continuous inference, real-time reasoning, and the ability to learn and adapt from ever-changing inputs. This is an important inflection point — a return to where software and hardware, or the virtual and physical worlds, meet. These systems must synthesize multimodal data from vision, language, audio, and sensor streams — sometimes simultaneously — and make context-aware decisions. Training these models will increasingly rely on synthetic data generated through simulation platforms like NVIDIA Omniverse, while inference will shift toward edge devices and new AI-native chips. This means an even greater capital expenditure boom in global data center infrastructure than previously anticipated. Huang stated that “the trillion-dollar industry of data centers will double in the next four to five years,” forecasting a staggering \$1 trillion in annual data center spend by 2028. While some fear that technologies like DeepSeek and efficient model compression

might reduce compute needs, the shift toward embodied AI — with its relentless demand for reasoning, adaptation, and real-world interaction — will far outweigh those savings. If anything, we're still underestimating what's needed.

With this transformation comes both promise and peril. On one hand, embodied AI will deliver extraordinary efficiency gains and unlock new sources of economic growth. Companies will see profit margins expand as robots and autonomous agents reduce labor costs, increase uptime, and optimize processes. On the other hand, the speed of deployment could outpace the speed of adaptation. As humanoids replace physical labor, and AI-native devices reduce the need for clerical and customer service roles, there will be growing concerns about job displacement, the erosion of middle-skill work, and societal inequality. The shift will require a major rethinking of workforce training, education systems, and economic safety nets.

There's also a psychological and cultural adjustment ahead. The presence of embodied hardware — from self-driving cars and delivery drones to robots assisting the elderly — will challenge our ideas of trust, safety, and human interaction. People may welcome the convenience and novelty, but many will also feel uneasy about intelligent machines becoming part of daily life. Like every industrial revolution before it, the embodiment of AI will bring anxiety alongside progress. Policymakers, business leaders, and technologists will need to work together to ensure the transition is inclusive, ethical, and human-centered.

In sum, we are entering a new era of AI — one where the intelligence is not abstract, but real, tangible, and increasingly autonomous. With Jensen Huang's announcement and the growing focus from industry leaders like Elon Musk, the groundwork is being laid for a future where AI walks, drives, assists, and collaborates with us in the real world. As AI gains embodiment, it also gains agency — and with it, the potential to transform society more deeply than any software revolution that has come before. The opportunities are immense but all of this comes at a time when we are distracted on what the tariff rates will be on April 2nd. It's time to focus on the robots.

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